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Disentangling the effects of perceived personal and group ethnic discrimination among secondary school students: The protective role of teacher–student relationship quality and school climate

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Abstract
Guided by risk and resilience and attachment perspectives, the present study examined whether teacher–student relationship quality and school climate can buffer against the deleterious effects of perceived personal and group ethnic discrimination on psychological and academic domains. We conducted multilevel analyses of seventh graders (40 classrooms; $N = 456$; 47% female) with different cultural self-identifications in Germany. Partially confirming pre-registered hypotheses, results indicated that high levels of perceived personal discrimination were negatively associated with global self-esteem and emotional school engagement. Contrary to our expectations, neither perceived personal nor group discrimination negatively predicted academic self-concept. In addition, teacher–student relationship quality but not school climate buffered the relationship between both personal and group discrimination and global self-esteem and emotional school engagement such that the association was less negative when relationship quality was high. Taken together, our results underscore the importance of considering the different targets of discrimination (i.e., personal self and own group), and

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that positive teacher–student relationship can be especially beneficial and empowering for youth who are exposed to ethnic discrimination.

KEYWORDS
academic self-concept, ethnic discrimination, school climate, school engagement, self-esteem, teacher–student relationship

1 | INTRODUCTION

Germany continues to be the most popular immigration destination in Europe (OECD, 2019), with a dynamically changing demographic landscape. Data available from the Federal Anti-Discrimination Agency (ADS, 2017) show that 31.4% of individuals in Germany report that they have experienced discrimination on the basis of ethnicity, gender, disability, religion, age, or sexual identity in the last 2 years. Among 18,000 surveyed participants, discrimination due to ethnic origin represented the second most prevalent experience, preceded by gender-based discrimination. According to a newer report published by the same agency (ADS, 2020), the number of ethnic discrimination episodes rose by almost 10%, representing the largest share for the first time (33% of the total cases). This is a worrying development, as these trends reflect that ethnic discrimination has increased in a significant way in Germany.

Ethnic discrimination can be defined as the differential treatment on the basis of ethnicity that disadvantages one or more members of an ethnic group (Quillian, 1995). In a national study on discrimination based on ethnicity (SVR-Forschungsbereich, 2018), more than one fifth (22.9%) of individuals of Turkish origin reported to have been strongly disadvantaged because of their ethnic belonging throughout the last 5 years. Although individuals of Turkish origin are often the primary target of discrimination (Moffitt, Juang, & Syed, 2019; Tucci, Eisnecker, & Brücker, 2014; Uslucan, 2017), also other well-established ethnic groups face discrimination because of their family heritage (Brenick, Titzmann, Michel, & Silbereisen, 2012; Gomolla, 2017). Yet there is scant research on the perceptions and the downstream consequences of ethnic discrimination on young adolescents in Germany. For ethnic minority youth in particular, being a victim of discriminatory actions is negatively associated with psychological well-being and academic achievement (Benner et al., 2018). In the present study, we investigate the effects of perceived ethnic discrimination on personal (i.e., global self-esteem) and academic domains (i.e., academic self-concept and emotional school engagement), and test for the buffering effects of teacher–student relationship quality and school climate in a sample of ethnic minority adolescents with different cultural self-identifications in Germany.

1.1 | The effects of ethnic discrimination on adolescents

A recent meta-analysis examining the effects of ethnic discrimination experiences on academic outcomes among adolescents has shown school engagement to be the most affected (Benner et al., 2018). School engagement is a robust predictor of academic achievement and school completion (Finn & Zimmer, 2013). Fredricks et al. (2004) describe three dimensions of this construct: behavioral (i.e., students' complying with school rules), cognitive (i.e., students' investment in learning activities), and emotional (i.e., students'

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positive and negative emotions attached to school, teachers, and classmates). The latter dimension is particularly susceptible to discrimination experiences because being disrespected or mistreated may be associated with negative emotions (Joseph, Peterson, Gordon, & Kamarck, 2020). Accordingly, research among ethnic minorities students in the United States (Li & Learner, 2011) identified steeper declines of emotional engagement compared to behavioral and cognitive engagement in response to ethnic discrimination.

Being exposed to ethnic discrimination, in which one's ethnic group is devalued, can represent a significant threat to one's self-esteem (Schmitt, Branscombe, Postmes, & Garcia, 2014). This could be true especially for early adolescents, who undergo identity changes that are directly related to self-concept (Zeiders, Umana-Taylor, & Derlan, 2013). Developmental researchers and social psychologists have long been sensitive to how discriminatory experiences can contribute to lowering global self-esteem, which in turn can have profound consequences for individuals' psychological functioning (Umaña-Taylor & Updegraff, 2007; Verkuyten, 1998). To date, however, the bulk of research on the effects of ethnic discrimination has focused on global self-esteem (Schmitt et al., 2014), leading to calls for greater attention to domain-specific self-concepts. Whereas self-esteem describes the global perception of the self as a person (Rosenberg, 1965), academic self-concept reflects a student's self-evaluation regarding a specific domain or ability in academic areas (Dickhäuser, Schöne, Spinath, & Stiensmeier-Pelster, 2002). Global self-esteem and academic self-concept are reciprocally associated, but only the latter consistently informs later achievement and educational attainment (Marsh & O'Mara, 2008). Thus, it is important distinguishing these constructs when addressing their relationship with ethnic discrimination.

1.2 | Perceived personal and group ethnic discrimination

In the literature on racism and discrimination (Jones, 1997), an important distinction has been made regarding the characteristics of the target. Whereas personal discrimination is directed at the individual self as member of a particular social group, group discrimination concentrates on a group as a whole (Verkuyten, Thijs, & Gharaei, 2019). Three decades ago, Taylor et al. (1990) argued that individuals tend to perceive higher levels of discrimination directed at their ethnic group than at themselves, a phenomenon called personal/group-discrimination discrepancy. More recently, meta-analytic findings (Schmitt et al., 2014) reported that the personal-group difference was larger for adolescent samples compared to adult samples. Hence, understanding how perceived personal and group discrimination are uniquely associated with different psychological and academic outcomes is particularly relevant for optimizing the development of ethnic minority youth.

With a few notable exceptions (Armenta & Hunt, 2009; Motti-Stefanidi & Asendorpf, 2012; Stevens & Thijs, 2018), research on ethnic discrimination with young adolescents has mainly assessed the detrimental effects of perceived personal discrimination. Following the personal/group-discrimination discrepancy hypothesis, research on youth showed that awareness of fellow group members being discriminated against (i.e., perceived group discrimination) is higher than discrimination against the self (Stevens & Thijs, 2018). However, whether personal discrimination is more harmful than perceived group discrimination remains unclear. A study conducted by Armenta and Hunt (2009) among Latino adolescents found contrasting patterns: whereas personal discrimination was negatively associated with self-esteem, group discrimination was positively correlated. Studies conducted with student samples showed that personal discrimination was associated with lowered self-esteem and diminished academic self-concept (Cogburn, Chavous, & Griffin, 2011; Wong, Eccles, & Sameroff, 2003). Stevens and Thijs (2018) found that higher levels of

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group discrimination were associated with more frequent internalizing and externalizing problems but not with lower self-esteem. Thus, there are some inconsistencies between the effects of perceived personal and group discrimination on psychological adjustment, which deserve further investigation.

1.3 | Guiding theoretical frameworks

This study is guided by two theoretical frameworks, the risk and resilience perspective (Masten & Coatsworth, 1998), and the attachment and relational perspective (Hamre & Pianta, 2001; Pianta, Hamre, & Allen, 2012). According to the risk and resilience perspective, risk factors are theorized to have a direct effect on maladjustment (Masten, 2001). Given the well-documented negative consequences of ethnic discrimination, there is consensus that discrimination represents a risk factor for ethnic minority youth. At the same time, children and adolescents have assets in their lives that can shield them against the negative consequences of known risk factors, by lessening the strength of the links between the risk factor and the negative outcome (Umaña-Taylor, 2016). As a result, young people can develop successfully despite adverse experiences and this interaction between risk and protective factors has been coined resilience (Southwick, Bonanno, Mastern, Panter-Brick, & Yehuda, 2014). Thus, investigating protective factors that shield against these negative experiences has important societal and educational implications.

Several studies have adopted the risk and resilience perspective, examining individual, peer, and family resources and their protective role in the association between perceived discrimination and negative adjustment outcomes for youth and adolescents (Delgado, Nair, Updegraff, & Umaña-Taylor, 2019; Doğan & Strohmeier, 2020; Moffitt et al., 2019; Motti-Stefanidi & Masten, 2013). At the individual level, for example, ethnic–racial identity can decrease vulnerabilities associated with discrimination, by supporting a sense of self despite belonging to stigmatized groups (for a meta-analysis see Yip, Wang, Mootoo, & Mirpuri, 2019). Family support is also an important protective factor. A supportive family alleviates the effects of discrimination by providing a source of immediate help in the face of stressful events (Hughes, 2003; Juang & Alvarez, 2010). Considerably less attention has been paid on assets or social protective factors youth may find in a key developmental context as school.

Complementing the risk and resilience perspective, the attachment perspective identifies positive relationships as an important asset for vulnerable youngsters, putting emphasis on the quality of relationships with significant non-parental figures such as teachers (Hamre & Pianta, 2001). Teachers represent secondary attachment figures that provide attachment-related safety and trust (Pianta et al., 2012). In addition, teachers' ability to define and negotiate sets of norms, goals, and values with their students contributes to the perceptions of overall school climate (Thapa, Cohen, Guffey, & Higgins-D'Alessandro, 2013). Accordingly, following research that has shown that the quality of relationship can be particularity beneficial for vulnerable youngsters (Roorda, Koomen, Spilt, & Oort, 2011), we theorized that assets in schools (i.e., teacher–student relationship quality and perceptions of school climate) may interact with risk factors, and thus mitigate negative effects of discriminatory experiences.

1.4 | Teacher–student relationship quality

For their proximity to students, child–adult relationships are central to children's emotional and cognitive development (Pianta et al., 2012). Teachers represent one caring

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adult that may contribute shaping children's relational models and subsequent behaviors (Sabol & Pianta, 2012). In secondary schools, however, teacher–student relationship quality tends to decline, the learning atmosphere becomes more formal, and students interact with greater numbers of teachers, making it more difficult to create a close bond (Roorda et al., 2011). A study among seventh graders (De Laet, Colpin, Goossens, Van Leeuwen, & Verschueren, 2014) found that students seldom rely on teachers when they are threatened, distressed, or upset. Nevertheless, young adolescents' perceptions of even one teacher being caring and supportive have been shown to be positively related with school achievement (Wentzel, 2002). These teacher characteristics have been rarely studied in relation to challenging situations such as discrimination through the eyes of early adolescents.

As ethnic minority students are more likely to experience discrimination in and outside of school, a positive teacher–student relationship functions as a protective factor (Göbel & Preusche, 2019; Juang et al., 2018). When adolescents feel devalued due to ethnic discrimination, caring teachers may offer a secure base and help students to feel connected with the school (Motti-Stefanidi & Masten, 2013). Among newly arrived immigrant students, perceiving strong bonds with teachers was positively linked to their school engagement over a 3-year period (Green, Rhodes, Hirsch, Suárez-Orozco, & Camic, 2008). However, recent research shows that caring relationships with teachers may not be enough to limit the consequences of discrimination among students who are exposed to high levels of discrimination (Price, Hill, Liang, & Perella, 2019). Moreover, it is also possible that low teacher–student relationship quality may exacerbate the negative consequences of discrimination, as misunderstandings and interpersonal conflicts with teachers could put an additional burden on those students (Makarova, 't Gilde, & Birman, 2019).

Although perceptions of interpersonal relationships originate within the person, they develop from exposure to intense situations that converge on consensual collective levels (Kozlowski & Ilgen, 2006). For example, students embedded in the same classrooms report greater similarity in perceptions of the affective relationship with their teachers than students who are enrolled in different classrooms (Spilt, Koomen, & Jak, 2012). Notably, few studies have empirically examined teacher–student relationship quality applying a multilevel approach at the classroom level (Maulana, Opdenakker, & Bosker, 2014; Spilt et al., 2012). In secondary schools, teachers teach multiple classes, and students are taught by multiple teachers. To obtain a precise estimate of teacher–student relationship quality, we examined student perceptions of relationship involving only homeroom teachers. Homeroom teachers spend most of the time with students teaching usually one or two subjects, and are also responsible for other issues such as counselling.

1.5 | Perceptions of school climate

School climate is a multidimensional construct, embracing the collective perceptions of students, teachers, and parents concerning organizational, instructional, and interpersonal domains of the learning environment (Thapa et al., 2013). One subdimension encompassed within the interpersonal domain includes the extent to which students feel emotionally safe in school, a high degree of relational trust, and the presence of clearly negotiated and defined rules (Thapa et al., 2013). Research shows that for early adolescents, positive interpersonal school climate (i.e., perceiving good relations with teachers and students) is positively associated with high self-esteem (Jia et al., 2009), academic self-concept (Scherer & Nilsen, 2016), and school engagement (Reyes, Brackett, Rivers, White, & Salovey, 2012).

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Previous research found that a positive school climate characterized by cooperation, cohesion, and mutual trust can protect against ethnic discrimination (Bayram et al., 2014; Byrd & Carter Andrews, 2016; Griffin, Cooper, Metzger, Golden, & White, 2017). In practice, this can help not only to create a more equitable school climate, but is also linked to school success and resilience among children at risk of negative academic outcomes (Bondy, Ross, Gallingane, & Hambacher, 2007). Our study brings attention to one indicator of school climate that explores students' perceptions of feeling valued and respected by other peers and teachers. As teacher–student relationship and school climate may be differently associated with student adjustment, it is important to explore their role separately.

1.6 | The current study

With the current research, we aim to address some important limitations in the body of knowledge regarding the effects of ethnic discrimination on ethnic minority youth. First, although extensive research has been carried out on ethnic discrimination in the United States, very few studies exist in German speaking countries (e.g., Brenick, Schachner, & Jugert, 2018; Doğan & Strohmeier, 2020), especially among early adolescents. Second, focusing on different targets of discrimination (i.e., perceived personal and group discrimination) this study aims at describing their unique relations with personal (i.e., global self-esteem) and academic domains (i.e., academic self-concept and emotional school engagement). Third, by simultaneously weighing the value of each predictor and accounting for the nested nature of the data, this investigation provides a rare opportunity to test the multilevel buffering effects of proximal interpersonal relationships (i.e., teacher–student relationship quality and school climate) at both individual and classroom level.

We formulated the following hypotheses (all of which were pre-registered and can be found on the Open Science Framework (OSF) project page, [https://osf.io/nxzs6]).

1.6.1 | Hypothesis 1

Following abundant evidence regarding the negative effects of ethnic discrimination among adolescents (see for a review Benner et al., 2018), we expected that greater perceived personal and group discrimination are negatively associated with self-esteem, academic self-concept, and emotional school engagement. Due to some conflicting empirical findings (Armenta et al., 2009; Stevens & Thijs, 2018; Wong et al., 2003), we examined the potentially differential role of perceived personal and group discrimination in an exploratory fashion.

1.6.2 | Hypothesis 2

In line with the protective factors literature (Motti-Stefanidi & Masten, 2013), the attachment perspective (Juang et al., 2018; Pianta et al., 2012), and school climate research (Thapa et al., 2013), we anticipated that teacher–student relationship quality and school climate may buffer the link between perceived personal and group discrimination and the three outcomes (i.e., global self-esteem, academic self-concept, and emotional school engagement), such that their association is weaker when teacher–student relationship quality is high and school climate is positive.

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2 | METHOD

2.1 | Participants and procedure

The sample included 456 seventh graders (47% female, mostly aged 12–13 years[1], 21.3% older than 13) from 40 classrooms in 12 schools in North Rhine-Westphalia, Germany. Data for this study were drawn from a sample of 828 adolescents who were recruited from a larger international cooperation focusing on hidden school dropout of students in different European countries, including Germany and Israel. Due to our interest in perceived ethnic discrimination of ethnic minority students, and as the term "immigrant background" does not capture discrimination based on ethnic group membership (Vietze, Moffitt, Schwarzenthal, & Civitillo, 2021), we focused on adolescents who self-identified with at least one other culture in addition to the German one. We assessed participants' cultural identification with an open response item: "*In Germany, people from different cultures (e.g., German, Turkish, Polish, Russian background) are living together. We would like to know, which culture or cultures do you mainly feel part of*?" We grouped participants into two different groups: (1) *only German identification*, (2) *another culture* or *two other cultures*. Three-hundred and seventy-two of the original group of 828 students were excluded because they identified only with German culture. The final sample of 456 adolescents reported more than 50 different cultural identifications, with Turkey, Poland, Italy, and Russia as the biggest heritage groups being represented.

Participating schools represented the different secondary schools tracks in this area, with 52% of students attending integrated secondary schools (*Gesamtschule*, which allow for either academic or vocational school-leaving certificates), 5.3% attending academic track schools (*Gymnasium*), and 10.7% low (*Hauptschule*) and 32% medium (*Realschule*) vocational track schools both allowing an apprenticeship upon graduation. After obtaining permission from school principals and parental consent, the questionnaire was administered during class time.

2.2 | Measures

The complete list of items for each construct in German and English can be found on the OSF project page (https://osf.io/nxzs6).

2.2.1 | Perceived personal and group discrimination

Eight items were used to assess perceived discrimination (four items for personal discrimination; e.g., "*I have been teased and insulted because of my cultural background*," $\omega = .76$; and four items for group discrimination; e.g., "*People from my culture have been threatened or attacked by people with another cultural background*," $\omega = .82$). The response scale ranged from 1 (strongly disagree) to 4 (strongly agree). Both subscales were adapted from Oppedal, Røysamb, and Heyerdahl (2005).

2.2.2 | Emotional school engagement

Emotional school engagement (six items; e.g., "*I feel excited by the work in school*," $\omega = .84$) was measured by a scale adapted from Fredricks et al. (2005), ranging from 1 (rarely) to 4 (always).

[1] Following data protection rules in this area, we were unable to ask participants the exact age.

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2.2.3 | Self-esteem

Self-esteem was measured by an adapted version of the Rosenberg's Self-Esteem Scale (Rosenberg, 1965). This scale incorporates two trait factors of self-esteem (positive and negative), comprising positively and negatively worded items (e.g., "*On the whole, I am satisfied with myself*," $\omega = .84$: e.g., "*All in all, I am inclined to think that I am a failure*," $\omega = .82$). The response scale ranged from 1 (strongly disagree) to 5 (strongly agree).

2.2.4 | Academic self-concept

In the academic self-concept scale developed in the German context (Dickhäuser et al., 2002), students were asked the extent to which they agreed with these statements applied to themselves using a 5-point Likert scale (five items, e.g., "*I'm very talented at school*," $\omega = .87$).

2.2.5 | Teacher–student relationship quality

We measured the quality of teacher–student relationship with a 4-point Likert scale by Schönbächler (2008) (five items, e.g., "*I have a good relationship with my homeroom teacher*"; "*My homeroom teacher is a role model for me*," $\omega = .86$), which is frequently used in the German context. The response scale ranged from 1 (strongly disagree) to 4 (strongly agree). High scores indicate a positive relationship with the homeroom teacher.

2.2.6 | School climate

We assessed school climate with an adapted scale of Moos and Tricket (1987) that comprised ten items (e.g., "*Student and teachers count on each other*," $\omega = .87$). The response scale ranged from 1 (strongly disagree) to 5 (strongly agree).

2.2.7 | Controls

We assessed gender with one item ("*What is your gender?*") with the response options (0) male and (1) female. As an indicator of family's cultural capital, we created a mean score by aggregating the following items: "*How many books do you have at home?*"; with few (0) or many books (1) responses, and "*Do you have your own desk at home where you can make your homework?*"; *Which of the following things do you have at home? Musical instruments, a calm place to study, classical literature, art work*, with no (0) or yes (1) responses. Finally, parents' occupational status (Highest International Socio-Economic Index of Occupational Status, HISEI; Ganzeboom, De Graaf, & Treiman, 1992) was assessed with an open question: *Which profession does your mother and father have?* This index ranges from 16 to 90 (higher scores indicate upper-level occupations, e.g., judges). We coded the higher of the two parental values in the analyses.

2.3 | Plan of data analysis

As preliminary analyses, we performed Confirmatory Factor Analysis (CFA) in *R* (4.0.0) using *lavaan* (Rosseel, 2012; 0.6–5) to assess the factor structure of all scales included in our study, using the items specified in the scales' original publication. Multiple indices were

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used to test overall model fits with the following indicating acceptable fit: comparative fit index (CFI) ≥ .90, Tucker–Lewis index (TLI) ≥ .90, root mean square error of approximation (RMSEA) ≤ .08, and standardized root mean square residual (SRMR) ≤ .08 (Marsh, Hau, & Wen, 2004). The Maximum Likelihood Robust (MLR) estimator was selected accounting for complete and incomplete data. We chose MLR estimator for its accuracy in estimating factor loadings with slightly non-normal distributed data (Li, 2016).

All scales but one (i.e., school climate) had good or excellent fit. For school climate, the initial one-factor model that included a single factor fitted the data poorly, $\chi^2(35) = 222.72$, $p = .00$, RMSEA = .11, 90% CI = [.10, .12]; CFI = .84; TLI = .79, and SRMR = .07. The two-factor model fitted the data significantly better than the one-factor model $\chi^2(34) = 93.82$, $p = .00$, RMSEA = .08, 90% CI = [.06, .09]; CFI = .95; TLI = .93, and SRMR = .06. Inspecting the two-factor solution, six items comprised perceptions of teacher–student interactions concerning common decisions in school (e.g., "*Part of school rules are decided by teachers' as well as students' votes*"; "*Teachers allow their students to express their views on decisions related to school rules*," $\omega = .84$), and four items pointed to perceptions of students-students interactions (e.g., "*Students count on each other*"; "*Students help each other even if they are not friends*," $\omega = .81$). Since the scale was multi-dimensional, both subscales and their buffering effects should have been examined separately. To avoid inflating Type I error rate, and thus containing the number of tests being conducted, we retained the first scale (i.e., six items), referring to perceptions of teacher–student interactions. See Supporting Information for the full list of items that were included (and excluded) in the current study (https://osf.io/nxzs6).

To account for the nested data structure—Level 1 students nested in Level 2 classrooms—we ran multilevel models in MPlus 8.3 (Muthén & Muthén, 2011–2018). We regressed each outcome variable (i.e., global self-esteem, academic self-concept, and emotional school engagement) on perceived personal and group discrimination. We examined the moderating role of teacher–student relationship quality and school climate at Level 1 and Level 2, controlling for gender, parents' occupational status, and family's cultural capital. We used a latent centering approach (Asparouhov & Muthén, 2019), by modeling the within and between-person components of an observed variable as latent variables. The latent mean centering approach is superior to observed centering because it accounts for the sampling error in the mean estimate. The intraclass correlations coefficients (ICC) were calculated for each outcome variable (i.e., self-esteem, emotional school engagement, and academic self-concept) and school context moderators (i.e., teacher–student relationship quality and school climate). All predictors at the Level 1 and Level 2 were grand-mean centered prior to the analyses. Missing data were handled in the multilevel models using Full-Information Maximum Likelihood (FIML) estimation, which uses all available data points to estimate parameters but excludes cases with missing values on covariates.

3 | RESULTS

3.1 | Preliminary analyses: Descriptives and bivariate correlations

Table 1 shows descriptives and bivariate correlations for all study variables. The ratio of missingness ranged between 0.9% and 9.1% across the variables. There was a strong positive correlation between perceived personal and group discrimination ($r = .64$, $p < .001$). We tested for multicollinearity (variance inflation factors, VIF) by regressing each outcome variable on personal discrimination, group discrimination, teacher–student relationship, school climate, gender, and family's cultural capital, and HISEI, using OLS

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TABLE 1 Descriptives and bivariate correlations between individual study variables

		Range	*M(SD)*	1	2	3	4	5	6	7	8	9	10
1	PPD	1–4	1.50(0.57)	—									
2	PGD	1–4	1.61(0.69)	.64***	—								
3	Emotional school engagement	1–4	2.46(0.74)	−.10**	−.08	—							
4	Global self-esteem	1–5	3.71(0.83)	−.23***	−.09	.36***	—						
5	Academic self-concept	1–5	3.31(0.91)	−.08	−.06	.41***	.42***	—					
6	Teacher–student relationship	1–4	2.92(0.76)	−.16***	−.08	.47***	.20***	.23***	—				
7	School climate	1–5	3.24(0.90)	−.16**	−.10*	.47***	.29***	.27***	.55***	—			
8	HISEI	17–89	44.81(14.3)	.09	.03	.12**	.17	.08	−.08	−.11*	—		
9	Family's cultural capital	0–1	0.69(0.23)	.06	.07	.22***	.03	.16***	.13**	.02	.22***	—	
10	Gender	0–1	—	−.05**	.00	.04	−.13**	−.11*	.03	−.02	.19**	.22***	—

Note. PPD, perceived personal discrimination; PPG, perceived group discrimination. Gender (female = 0, male = 1).
*$p < .05$.
**$p < .01$.
***$p < .001$. $N = 456$.

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regression. These analyses showed VIF values of 1.615 for group discrimination and 1.654 for personal discrimination when including them in the same model, well within the acceptable range of below 5 (Belsley, Kuh, & Welsch, 2005). However, inclusion of both personal and group discrimination in the same model led to nonignorable increases (>.02) in the standard errors of the regression coefficients. In addition, while the zero-order relationship between self-esteem and group discrimination was negative it changed sign when including it in the same model with group discrimination, suggestive of multicollinearity. We thus decided to run analyses separately for personal and group discrimination. In line with previous research, the mean score of perceived group discrimination (M = 1.61, SD = 0.69) was higher than that of perceived personal discrimination (M = 1.50, SD = 0.57), M_{Diff} = −.11 [−.16, −.06], $t(448) = -4.182$, p <.001. Perceived personal discrimination was negatively related with emotional school engagement and global self-esteem, while no association with academic self-concept was found. Although coefficients were negative, perceived group discrimination did not correlate significantly with the three outcomes. Next, perceived personal discrimination was negatively related to teacher–student relationship quality and school climate, while group discrimination was only negatively related to school climate. Finally, all three outcomes were positively related.

3.2 | Multilevel models

To test our hypotheses, we conducted multilevel analyses using a latent centering approach. In a first step, ICCs were calculated for each outcome variable. The ICCs for emotional school engagement, global self-esteem, and academic school achievement were 0.07, 0.02, 0.04, respectively, suggesting medium to little variance at the classroom level (see Supporting Information for the ICCs of all other variables). For each outcome variable, we ran separate multilevel analyses for personal and group discrimination because of the multicollinearity issue noted above. In Model 1, personal or group discrimination were entered next to control variables (gender, HISEI, and family cultural capital) as predictors at the individual level (testing Hypothesis 1). In model 2, we included teacher–student relationship quality and school climate and respective two-way interactions between both moderators and individual or group discrimination at level 1 (testing Hypothesis 2). In model 3, both moderators were also included at level 2 and cross-level interactions (i.e., involving group-level moderators) were tested for personal or group discrimination.

Partly in line with Hypothesis 1, perceived personal but not group discrimination was significantly and negatively associated with emotional school engagement (see Table 2, Model 1). In support of Hypothesis 2, there was a significant group discrimination × teacher–student relationship quality interaction (b = .14, p = .011) at the individual level (see Table 3, Model 2). To better understand this interaction, we plotted it by using the Johnson–Neyman (JN) procedure (Johnson & Neyman, 1936). The JN procedure allows to identify the value(s) of a continuous moderator that determines where the conditional effect of an independent variable on a dependent variable transition from statistically non-significant to significant (Hayes, 2013). As shown in Figure 1, the effect of group discrimination on emotional school engagement was significant and negative at values below −0.55 of teacher–student relationship quality (equivalent to 0.71 SD below the mean). As teacher–student relationship quality improved the association between group discrimination and emotional school engagement was no longer significant, indicating a buffering effect. In addition, there was a significant group discrimination × school climate interaction (b = −.12, p = .006) at the individual level. As shown in Figure 2, the effect of group

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TABLE 2 Multilevel model predicting emotional school engagement from personal discrimination

	Model 1 ($n = 394$)		Model 2 ($n = 379$)		Model 3 ($n = 379$)	
	B	*SE*	*B*	*SE*	*B*	*SE*
Level 1—Individual						
Gender	.02	.04	.05	.05	.05	.04
Family's cultural capital	.26***	.06	.53**	.16	.47*	.17
HISEI	.08	.05	.01***	.00	.01***	.00
PPD	−.17*	.08	−.02	.06	—	—
Teacher–student relationship quality			.29***	.07	.37***	.07
School climate			.31***	.04	.29***	.04
PPD × Teacher–student relationship quality			.07	.08	.17*	.08
PPD × School climate			−.07	.07	−.02	.08
Level 2—Classroom						
Teacher–student relationship quality					−.11	.11
School climate					.60***	.16
PPD × Teacher–student relationship quality					−.07	.33
PPD × School climate					−.45	.47

Note. PPD, perceived personal discrimination.
*$p < .05$.
**$p < .01$.
***$p < .001$. Unstandardized regression coefficients.

TABLE 3 Multilevel model predicting emotional school engagement from group discrimination

	Model 1 ($n = 394$)		Model 2 ($n = 379$)		Model 3 ($n = 379$)	
	B	*SE*	*B*	*SE*	*B*	*SE*
Level 1—Individual						
Gender	.03	.06	.06	.04	.04	.04
Family's cultural capital	.82***	.20	.50**	.15	.50**	.16
HISEI	.00	.00	.01***	.00	.01**	.00
PGD	−.08	.05	−.01	.04	–	–
Teacher–student relationship quality			.30***	.06	.39***	.06
School climate			.30***	.04	.29***	.04
PGD × Teacher–student relationship quality			.14*	.06	.14	.07
PGD × School climate			−.12*	.04	−.11*	.05
Level 2—Classroom						
Teacher–student relationship quality					−.05	.13
School climate					.48**	.18
PGD × Teacher–student relationship quality					.02	.16
PGD × School climate					−.05	.24

Note. PGD, perceived group discrimination.
*$p < .05$.
**$p < .01$.
***$p < .001$. Unstandardized regression coefficients.

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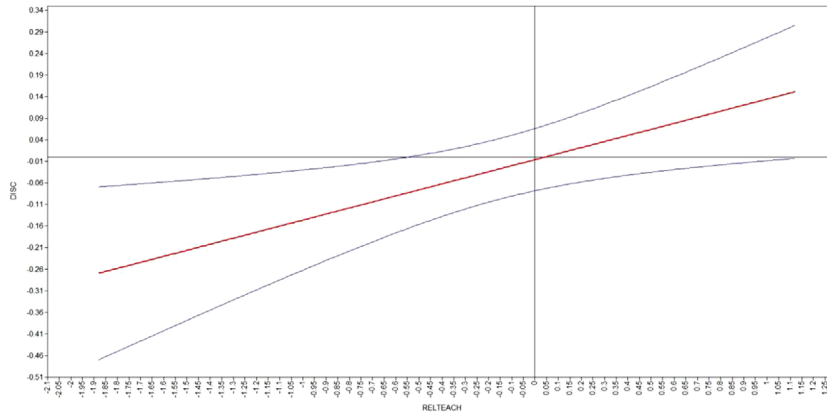

FIGURE 1 Johnson–Neyman plot for the adjusted effect of perceived group discrimination on emotional school engagement. The red line represents values of the adjusted effect as a function of student–teacher relationship values at the individual level. The blue lines represent 95% confidence bands around the adjusted effect

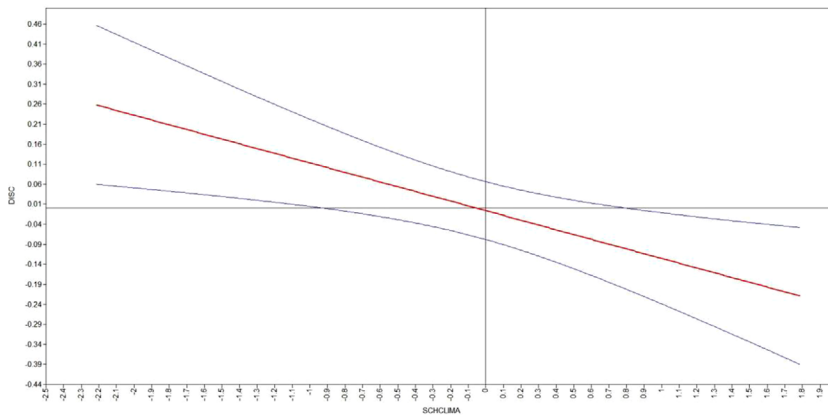

FIGURE 2 Johnson–Neyman plot for the adjusted effect of perceived group discrimination on emotional school engagement. The red line represents values of the adjusted effect as a function of perceived school climate at the individual level. The blue lines represent 95% confidence bands around the adjusted effect

discrimination on emotional school engagement was positive and significant when school climate was perceived to be low (lower than −0.9 from the mean) and it was negative and significant when school climate was perceived to be high (higher than 0.8 from the mean). Thus, instead of a buffering effect, a positive school climate seemed to have an aggravating effect. There were no significant cross-level interactions.

The multilevel models for global self-esteem (see Tables 4 and 5) showed that personal discrimination was significantly negatively associated with self-esteem while the association between group discrimination and self-esteem was negative but not significant. There were no significant interaction effects at the individual level. But there were two cross-level interactions, suggesting that teacher–student relationship quality at the classroom level moderated the association between both personal (b = .89, p = .054) and group discrimination (b = .62, p = .003) and self-esteem (depicted in Figures 3 and 4, respectively). As predicted, for students attending classrooms with low teacher–student relationship quality personal discrimination was significantly associated with lower self-esteem,

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TABLE 4 Multilevel model predicting global self-esteem from personal discrimination

	Model 1 ($n = 392$)		Model 2 ($n = 380$)		Model 3 ($n = 380$)	
	B	*SE*	*B*	*SE*	*B*	*SE*
Level 1—Individual						
Gender	−.31***	.08	−.30***	.08	−.32***	.09
Family's cultural capital	.17	.21	.09	.21	.07	.20
HISEI	.01	.00	.01**	.00	.01**	.00
PPD	−.34***	.10	−.26**	.08	—	—
Teacher–student relationship quality			.08	.06	.19*	.08
School climate			.24***	.05	.23***	.06
PPD × Teacher–student relationship quality			.05	.09	−.01	.10
PPD × School climate			.01	.10	.06	.13
Level 2—Classroom						
Teacher–student relationship quality					−.25	.15
School climate					.36	.24
PPD × Teacher–student relationship quality					.89+	.46
PPD × School climate					−.94	.68

Note. PPD, perceived personal discrimination.
+$p < .10$.
*$p < .05$.
**$p < .01$.
***$p < .001$. Unstandardized regression coefficients.

TABLE 5 Multilevel model predicting global self-esteem from group discrimination

	Model 1 ($n = 392$)		Model 2 ($n = 380$)		Model 3 ($n = 380$)	
	B	*SE*	*B*	*SE*	*B*	*SE*
Level 1—Individual						
Gender	−.30**	.09	−.27***	.09	−.30***	.09
Family's cultural capital	.15	.21	.01	.21	.03	.19
HISEI	.00	.00	.01*	.00	.01	.00
PGD	−.07	.06	−.05	.05	—	—
Teacher–student relationship quality			.11	.06	.22*	.09
School climate			.26***	.05	.25***	.06
PGD × Teacher–student relationship quality			.08	.08	−.07	.08
PGD × School climate			−.11	.06	−.05	.08
Level 2—Classroom						
Teacher–student relationship quality					−.31	.17
School climate					.42	.24
PGD × Teacher–student relationship quality					.62**	.21
PGD × School climate					−.46	.34

Note. PGD, perceived group discrimination.
*$p < .05$.
**$p < .01$.
***$p < .001$. Unstandardized regression coefficients.

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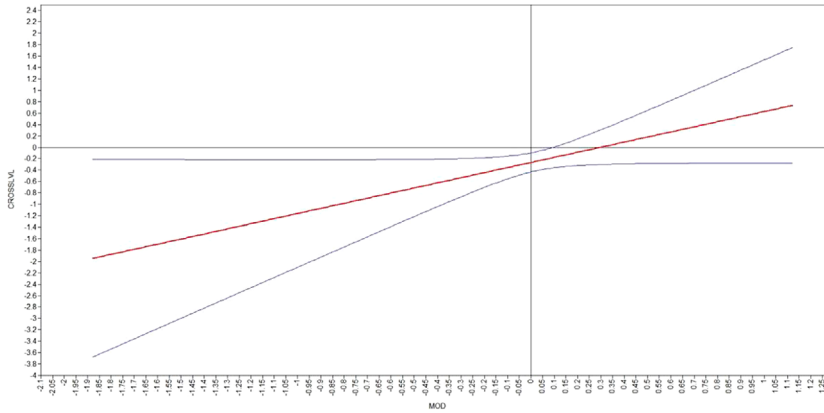

FIGURE 3 Johnson–Neyman plot for the adjusted effect of perceived personal discrimination on self-esteem. The red line represents values of the adjusted effect as a function of student–teacher relationship values at the classroom level. The blue lines represent 95% confidence bands around the adjusted effect

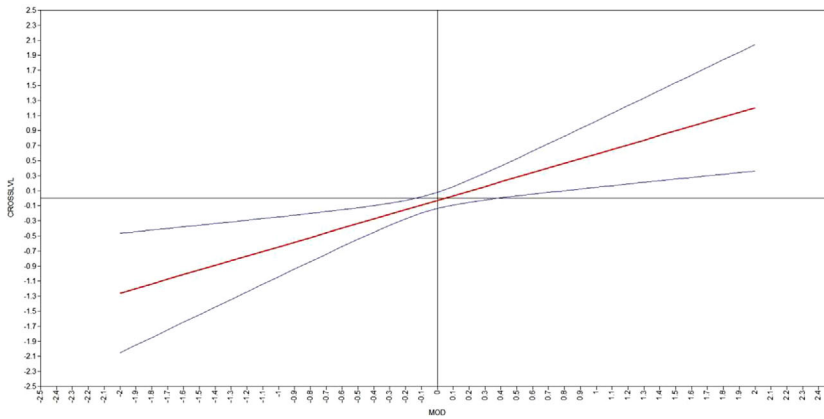

FIGURE 4 Johnson–Neyman plot for the adjusted effect of perceived group discrimination on self-esteem. The red line represents values of the adjusted effect as a function of student–teacher relationship values at the classroom level. The blue lines represent 95% confidence bands around the adjusted effect

while for students attending classrooms with slightly above average teacher–student relationship quality (above 0.1 from the mean) the association between personal discrimination and self-esteem was no longer significant. Similarly, for students attending classrooms with lower teacher–student relationship quality (lower than −0.1 from the mean) group discrimination was significantly associated with lower self-esteem. For students attending classroom with teacher–student relationship quality between −0.1 and 0.4 from the mean, group discrimination was not significantly associated with self-esteem, indicative of a buffering effect of the teacher–student relationship. Yet, unexpectedly, for students attending classrooms with teacher–student relationship quality of 0.4 or higher than the mean group discrimination was significantly and positively associated with self-esteem.

The multilevel models reported in Tables 6 and 7 showed that neither personal nor group discrimination were significantly linked with academic self-concept. There were also no significant individual-level or cross-sectional interactions.

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TABLE 6 Multilevel model predicting academic self-concept from personal discrimination

	Model 1 ($n = 388$)		Model 2 ($n = 378$)		Model 3 ($n = 378$)	
	B	*SE*	*B*	*SE*	*B*	*SE*
Level 1—Individual						
Gender	−.22**	.07	−.24**	.07	−.24***	.07
Family's cultural capital	.68**	.20	.51*	.23	.45	.26
HISEI	.00	.00	.01	.00	.01	.00
PPD	−.17	.11	−.08	.09	—	—
Teacher–student relationship quality			.18*	.08	.26**	.09
School climate			.22*	.08	.18	.09
PPD × Teacher–student relationship quality			.05	.09	.08	.10
PPD × School climate			.01	.11	−.01	.13
Level 2—Classroom						
Teacher–student relationship					−.36	.29
School climate					.81	.46
PPD × Teacher–student relationship quality					.21	.54
PPD × School climate					−.33	.89

Note. PPD, perceived personal discrimination.
*$p < .05$.
**$p < .01$.
***$p < .001$. Unstandardized regression coefficients.

TABLE 7 Multilevel model predicting academic self-concept from group discrimination

	Model 1 ($n = 388$)		Model 2 ($n = 378$)		Model 3 ($n = 378$)	
	B	*SE*	*B*	*SE*	*B*	*SE*
Level 1—Individual						
Gender	−.21**	.07	−.23***	.06	−.25***	.07
Family's cultural capital	.68**	.21	.45	.23	.46	.24
HISEI	.00	.00	.01	.00	.01	.00
PGD	−.08	.07	−.03	.07	—	—
Teacher–student relationship quality			.19*	.09	.26*	.10
School climate			.22*	.08	.19*	.09
PGD × Teacher–student relationship quality			.13	.08	.06	.11
PGD × School climate			−.14	.09	−.14	.11
Level 2—Classroom						
Teacher–student relationship					−.27	.22
School climate					.65	.34
PGD × Teacher–student relationship quality					.19	.24
PGD × School Climate					−.08	.42s

Note. PGD, perceived group discrimination.
*$p < .05$.
**$p < .01$.
***$p < .001$. Unstandardized regression coefficients.

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4 | DISCUSSION

Negative consequences of ethnic discrimination are well known for adults and adolescents. Yet, few studies have examined correlates of perceived personal and group discrimination simultaneously, exploring their relationships with psychological (i.e., global self-esteem) and academic domains (i.e., academic self-concept and emotional school engagement). In addition, there is a need to better understand the protective factors that shield against these negative experiences. Accordingly, building on risk and resilience and attachment perspectives, the second aim of the present study was to shed light on the moderating effects of two protective factors present in the school context (i.e., teacher–student relationship quality and school climate) at the individual and classroom level.

Personal discrimination (but not group discrimination) was consistently negatively related to the different outcomes. We found that teacher–student relationship quality but not school climate moderated the discrimination effects, depending on the type of outcome. Taken together, these cross-sectional findings provide preliminary but important evidence from which future research can build to further study the two levels of discrimination, namely perceived personal and group discrimination. In addition, these findings offer an opportunity for initial teaching preparation and professional development programs to acknowledge the importance of teacher–student relationship for youth who are exposed to ethnic discrimination. In particular, teacher empathy and perspective taking are crucial for building better student–teacher relationship (Civitillo & Juang, 2020), by improving the quality of intercultural student–teacher interactions, so that students feel they are understood and valued.

Although perceived group discrimination was higher than personal discrimination, only the latter form was consistently associated with lower emotional school engagement and lower global self-esteem, partially supporting our first hypothesis. Group discrimination was unrelated to emotional school engagement and academic self-concept. This implies that youth can give different importance or salience of the targets of discrimination (i.e., personal self and own group). Our study extends previous studies (Armenta & Hunt, 2009; Stevens & Thijs, 2018), by focusing on a highly diverse sample of participants rather than one ethnic minority group (i.e., Latinos and Dutch-Moroccans, respectively). The lack of associations between discrimination (regardless of the type) and academic self-concept might indicate that the internalization of discrimination and prejudice has not yet permeated the perceived competencies in the educational sphere. It could also be argued that these results are due to the fact that we assessed academic self-concept in general and not in specific subject domains (e.g., in German).

Our investigation showed that only personal discrimination was marginally negative correlated with emotional school engagement. This finding is partially in line with meta-analytical results that showed that school engagement is lowered by discrimination experiences (Benner et al., 2018). It is important to highlight that in Benner's and colleagues' meta-analysis school engagement was operationalized in a more general manner. Relatedly, this suggests that when addressing the relationship between discrimination and school engagement, it is important to assess its different dimensions (i.e., behavioral, cognitive, and emotional; Fredricks, Blumenfeld, & Paris, 2004), because they may change differently in response of discriminatory experiences (Motti-Stefanidi & Masten, 2013).

The second hypothesis pertained to the buffering effects of protective factors embedded in the school context. Based on protective factors literature (Motti-Stefanidi & Masten, 2013), the attachment perspective (Juang et al., 2018; Pianta et al., 2012), and school climate research (Thapa et al., 2013), we hypothesized that teacher–student relationship quality and school climate would buffer the link between perceived personal and

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group discrimination and the three outcomes, such that their association is weaker when teacher–student relationship quality is high and school climate is positive. This hypothesis was partially supported. We found consistent evidence for the buffering role of teacher–student relationship quality for two out of three outcomes (i.e., global self-esteem and emotional school engagement). These findings are congruent with other studies, indicating that high teacher–student relationship quality is protective for ethnic minority students (Green et al., 2008; Price et al., 2019). This means that, following attachment perspectives (Juang et al., 2018; Pianta et al., 2012), teachers may play an important role as sources of support for immigrant youth by providing attachment functions (sense of closeness, safety, and confidence). It is important to note that the vast majority of teachers in Germany (92.8%) do not have an immigrant background (Statistiches Bundesamt, 2016). Thus, schools are often suggested to transport societal norms to students (Schachner, Juang, Moffitt, & Van de Vijver, 2018). Providing a warm and close relationship with their students may thus be a welcome contrast to widespread negative societal and political discourse about immigrants in media coverage. Furthermore, our results show that, even though secondary students may feel less closed with their teachers when they grow older (De Laet et al., 2014; Roorda et al., 2011), for early adolescents, a positive relationship with non-parental figures such as teachers still represents an important protective factor.

Contrary to our expectations, school climate did not buffer against the deleterious effects of ethnic discrimination. A general school atmosphere of trust and respect has been linked to school success and resilience among ethnic minority students (Bondy et al., 2007). However, our results suggest that it may be insufficient to protect against discrimination. That is, adolescents may suffer from discrimination, even if they are placed in a school which promotes general respect and fairness between teachers and students. Another interpretation of this finding is that our measure of climate did not assess how cultural and ethnic diversity is dealt with at school. Previous studies that specifically investigated diversity climate found more consistent effects of its moderating role (Brenick et al., 2018; Byrd & Carter Andrews, 2016; Griffin et al., 2017).

Unexpectedly, group discrimination was associated with lower emotional school engagement when school climate was perceived to be high rather than low. One possible explanation is that the items assessing school climate tap into relations between students and teachers concerning common decisions in school, while the items assessing teacher–student relationship quality refer to the emotional bond of the student to their teacher. Perceiving that the teacher strives for an open school climate may not be sufficient when making experiences of ethnic discrimination because it does not deal with the specific problem of the student. A buffering effect may only occur when the student has a good and trusting relationship with the teacher.

Finally, we observed evidence for the personal-group discrimination discrepancy (Taylor, Wright, Moghaddam, & Lalonde, 1990), in which perceived group discrimination was higher than perceived personal discrimination. These results are in line with prior work examining this discrepancy in adults (Schmitt et al., 2014) as well as among ethnic minority adolescents (Armenta & Hunt, 2009; Motti-Stefanidi & Asendorpf, 2012; Stevens & Thijs, 2018). One explanation for this discrepancy concerns the denial of personal discrimination (Taylor, Wright, & Porter, 1994), that is individuals tend to feel relatively better compared to the dramatic examples of discrimination reported by other disadvantaged group members. In our sample, however, personal discrimination and group discrimination were positively related, suggesting that adolescents did not deny personal discrimination or exaggerate group discrimination. Rather, they tend to consider discrimination not as a solitary experience, but a common burden shared with in-group members.

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4.1 | Limitations and suggestions for future research

The results of this study should be interpreted with caution in light of some limitations. The first limitation refers to the cross-sectional design of the study. This makes hard to exclude possible reciprocal influences of discrimination and the outcomes under investigation in our study. In line with our theoretical predictions, however, longitudinal investigations suggest that perceived ethnic discrimination predicted changes in global self-esteem and academic self-concept growth across the high school years (Wong et al., 2003; Zeiders et al., 2013). A second limitation was the aggregation of participants with different cultural identifications into a pan-ethnic group. We are aware that in Germany, as in other countries, some ethnic minorities are more likely to face discrimination than others. As such, the effects of discrimination could be muted for not considering participants particular heritage (Vietze et al., 2021). A third limitation was the general nature of the personal and group discrimination items. If they were measured at the school level (e.g., ethnic discrimination occurring in school) this would perhaps make it easier to examine buffering roles of school factors (e.g., school climate). Nevertheless, discrimination can occur everywhere, thus we did not want to restrict the focus of such negative experiences to the school context.

Future research should continue assessing the unique associations between perceived personal and group discrimination with different psychological and academic outcomes. On a related note, it is warranted to distinguish global self-esteem and academic self-concept given the variation in their relationships with ethnic discrimination. For the latter, researchers need to increase the specificity of subject domain and explore possible variations across academic areas (e.g., general academic self-concept vs. German- or math-related self-concept). Next to consider the target of discrimination (i.e., personal self and own group), future studies could investigate the different forms that discrimination can take, for example open or subtle. In addition, an effort should be made to assess source-specific experiences of discrimination (e.g., peer based, teacher based, or combination of both). Different sources of discrimination might have differential consequences for ethnic minority students (Doğan & Strohmeier, 2020). To increase ecological validity of reported discrimination experiences, future research could make use of experience-sampling methods such as daily diaries. Daily diaries allow a rich understanding of the immediate impact of discrimination, exerting less cognitive effort for participants, and thus reducing the risk of retrospective bias in self-reported experiences (Yip, Cheon, & Erhardt, 2020).

5 | CONCLUSION

Ethnic discrimination is becoming widespread in Germany. However, still little is known about the detrimental consequences of discrimination for ethnic minority youth. By examining the different targets of discrimination (i.e., personal self and own group), we showed that they are differently related to psychological and academic domains. Our study highlights that a positive and trusting relationship between teachers and their students can function as a protective factor in the face of ethnic discrimination. To strengthen student's self-concept and prevent disengagement, strategies for the creation of a supportive school environment should become an important part in teacher education and professional development (Göbel & Frankenmölle, 2020). Pre- and in-service teachers should be encouraged and supported to focus on teacher–student relationship as well as on the prevention of discrimination.

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DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

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REFERENCES

Antidiskriminierungsstelle des Bundes (ADS). (2020). *Gleiche Rechte, gleiche Chancen - Jahresbericht der Antidiskriminierungsstelle des Bundes [Equal rights, equal opportunities - Annual report of the Federal Anti-Discrimination Agency]*. Antidiskriminierungsstelle des Bundes.

Antidiskriminierungsstelle desBundes (ADS). (2017). *Diskriminierungserfahrungen in Deutschland. Erste Ergebnisse einer repräsentativen Betroffenenbefragung [Experiences of discrimination in Germany: Initial results of a representative survey and a survey of the people affected.]*. Antidiskriminierungsstelle des Bundes.

Armenta, B. E., & Hunt, J. S. (2009). Responding to societal devaluation: Effects of perceived personal and group discrimination on the ethnic group identification and personal self-esteem of Latino/Latina adolescents. *Group Processes & Intergroup Relations*, *12*(1), 23–39.

Asparouhov, T., & Muthén, B. (2019). Latent variable centering of predictors and mediators in multilevel and time-series models. *Structural Equation Modeling: A Multidisciplinary Journal*, *26*(1), 119–142.

Bayram Özdemir, S., & Stattin, H. (2014). Why and when is ethnic harassment a risk for immigrant adolescents' school adjustment? Understanding the process and conditions. *Journal of Youth and Adolescence*, *43*, 1252–1265.

Belsley, D. A., Kuh, E., & Welsch, R. E. (2005). *Regression diagnostics: Identifying influential data and sources of collinearity*. Hoboken, NJ: John Wiley & Sons.

Benner, A. D., Wang, Y., Shen, Y., Boyle, A. E., Polk, R., & Cheng, Y. P. (2018). Racial/ethnic discrimination and well-being during adolescence: A meta-analytic review. *American Psychologist*, *73*, 855–883.

Bondy, E., Ross, D. D., Gallingane, C., & Hambacher, E. (2007). Creating environments of success and resilience: Culturally responsive classroom management and more. *Urban Education*, *42*(4), 326–348.

Brenick, A., Schachner, M. K., & Jugert, P. (2018). Help or hindrance? Minority versus majority cross-ethnic friendships altering discrimination experiences. *Journal of Applied Developmental Psychology*, *59*, 26–35.

Brenick, A., Titzmann, P. F., Michel, A., & Silbereisen, R. K. (2012). Perceptions of discrimination by young diaspora migrants: Individual- and school-level associations among adolescent ethnic German immigrants. *European Psychologist*, *17*, 105–199.

Byrd, C. M., & Andrews, D. J. C. (2016). Variations in students' perceived reasons for, sources of, and forms of in-school discrimination: A latent class analysis. *Journal of School Psychology*, *57*, 1–14.

Civitillo, S., & Juang, L. P. (2020). How to best prepare teachers for multicultural schools. In P. F. Titzmann & P. Jugert (Eds.), *Youth in superdiverse societies: Growing up with globalization, diversity, and acculturation* (pp. 285–301). Routledge.

Cogburn, C. D., Chavous, T. M., & Griffin, T. M. (2011). School-based racial and gender discrimination among African American adolescents: Exploring gender variation in frequency and implications for adjustment. *Race and Social Problems*, *3*(1), 25–37.

De Laet, S., Colpin, H., Goossens, L., Van Leeuwen, K., & Verschueren, K. (2014). Comparing parent–child and teacher–child relationships in early adolescence: Measurement invariance of perceived attachment-related dimensions. *Journal of Psychoeducational Assessment*, *32*(6), 521–532.

Delgado, M. Y., Nair, R. L., Updegraff, K. A., & Umaña-Taylor, A. J. (2019). Discrimination, parent-adolescents conflict, and peer intimacy: Examining risk and resilience in Mexican-origin youths' adjustment trajectories. *Child Development*, *90*(3), 894–910.

Dickhäuser, O., Schöne, C., Spinath, B., & Stiensmeier-Pelster, J. (2002). Die Skalen zum akademischen Selbstkonzept: Konstruktion und berprüfung eines neuen Instrumentes [The scale of Academic Self-Concept: Construction and validation of a new instrument]. *Zeitschrift für Differentielle und Diagnostische Psychologie*, *23*, 393–405.

Doğan, A., & Strohmeier, D. (2020). The role of discrimination, acculturation, and ethnic identity in predicting psychosocial functioning of Turkish immigrant youth. In D. Güngor & D. Strohmeier (Eds.), *Contextualizing immigrant and refugee resilience: Cultural and acculturation perspectives* (pp. 99–122). Springer.

15348687, 2021, 177, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/cad.20415, Wiley Online Library on [17/12/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Finn, J. D., & Zimmer, K. (2013). Student engagement: What is it? Why does it matter? In S. L. Chiristenson, A. L., Reschly, & C. Wylie (Eds), *Handbook of research on student engagement* (pp. 97–131). Springer.

Fredricks, J. A., Blumenfeld, P. C., & Paris, A. H. (2004). School engagement: Potential of the concept, state of the evidence. *Review of Educational Research*, *74*, 59–109.

Fredricks, J. A., Blumenfeld, P., Friedel, J., & Paris, A. (2005). School engagement. In K. A. Moore & L. H. Lippman (Eds.), *What do children need to flourish?* (pp. 305–321). Springer.

Ganzeboom, H. B. G., De Graaf, P. M., & Treiman, D. J. (1992). A standard international socio-economic index of occupational status. *Social Science Research*, *21*, 1–56.

Göbel, K., & Frankemölle, B. (2020). Interkulturalität und Wohlbefinden im Schulkontext. In T. Ringeisen, P. Genkova, & F. T. L. Leong (Eds.), *Handbuch Stress und Kultur* (pp. 1–17). Springer.

Göbel, K., & Preusche, Z. M. (2019). Emotional engagement among minority youth: The relevance of cultural identity, perceived discrimination, and perceived support. *Intercultural Education*, *30*(5), 547–563.

Gomolla, M. (2017). Direkte und indirekte, institutionelle und strukturelle Diskriminierung. In A. Scherr, A. El-Mafaalani, & G. Yüksel (Eds), *Handbuch Diskriminierung* (pp. 133–155). Springer: VS.

Green, G., Rhodes, J., Hirsch, A. H., Suárez-Orozco, C., & Camic, P. M. (2008). Supportive adult relationships and the academic engagement of Latin American immigrant youth. *Journal of School Psychology*, *46*(4), 393–412.

Griffin, C. B., Cooper, S. M., Metzger, I. W., Golden, A. R., & White, C. N. (2017). School racial climate and the academic achievement of African American high school students: The mediating role of school engagement. *Psychology in the Schools*, *54*(7), 673–688.

Hamre, B. K., & Pianta, R. C. (2001). Early teacher-child relationships and the trajectory of children's school outcomes through eighth grade. *Child Development*, *72*, 625–638.

Hayes, A. F. (2013). *Introduction to mediation, moderation, and conditional process analysis: A regression-based approach*. New York, NY: Guilford Press.

Hughes, D. (2003). Correlates of African American and Latino parents' messages to children about ethnicity and race: A comparative study of racial socialization. *American Journal of Community Psychology*, *31*(1-2), 15–33.

Jia, Y., Way, N., Ling, G., Yoshikawa, H., Chen, X., Hughes, D., Ke, X., & Lu, Z. (2009). The influence of student perceptions of school climate on socioemotional and academic adjustment: A comparison of Chinese and American adolescents. *Child Development*, *80*, 1514–1530.

Johnson, P. O., & Neyman, J. (1936). Tests of certain linear hypotheses and their applications to some educational problems. *Statistical Research Memoirs*, *1*, 57–93.

Jones, J. M. (1997). *Prejudice and racism*. McGraw-Hill Humanities, Social Sciences & World Languages.

Joseph, N. T., Peterson, L. M., Gordon, H., & Kamarck, T. W. (2020). The double burden of racial discrimination in daily-life moments: Increases in negative emotions and depletion of psychosocial resources among emerging adult African Americans. *Cultural Diversity and Ethnic Minority Psychology*.

Juang, L. P., & Alvarez, A. A. (2010). Discrimination and adjustment among Chinese American adolescents: Family conflict and family cohesion as vulnerability and protective factors. *American Journal of Public Health*, *100*(12), 2403–2409.

Juang, L. P., Simpson, J. A., Lee, R. M., Rothman, A. J., Titzmann, P. F., Schachner, M. K., Korn, L., Heinemeier, D., & Betsch, C. (2018). Using attachment and relational perspectives to understand adaptation and resilience among immigrant and refugee youth. *American Psychologist*, *73*(6), 797–811.

Kozlowski, S. W., & Ilgen, D. R. (2006). Enhancing the effectiveness of work groups and teams. *Psychological Science in the Public Interest*, *7*(3), 77–124.

Li, C. H. (2016). Confirmatory factor analysis with ordinal data: Comparing robust maximum likelihood and diagonally weighted least squares. *Behavior Research Methods*, *48*, 936–949.

Li, Y., & Lerner, R. M. (2011). Trajectories of school engagement during adolescence: Implications for grades, depression, delinquency, and substance use. *Developmental Psychology*, *47*(1), 233.

Makarova, E., 't Gilde, J., & Birman, D. (2019). Teachers as risk and resource factors in minority students' school adjustment: An integrative review of qualitative research on acculturation. *Intercultural Education*, *30*(5), 448–477.

Marsh, H. W., Hau, K.-T., & Wen, Z. (2004). In search of golden rules: Comment on hypothesis-testing approaches to setting cutoff values for fit indexes and dangers in overgeneralizing Hu and Bentler's (1999) findings. *Structural Equation Modeling: A Multidisciplinary Journal*, *11*, 320–341.

Marsh, H. W., & O'Mara, A. (2008). Reciprocal effects between academic self-concept, self-esteem, achievement, and attainment over seven adolescent years: Unidimensional and multidimensional perspectives of self-concept. *Personality and Social Psychology Bulletin*, *34*(4), 542–552.

Masten, A. S. (2001). Ordinary magic: Resilience processes in development. *American Psychologist*, *56*, 227–238.

Masten, A. S., & Coatsworth, J. D. (1998). The development of competence in favorable and unfavorable environments: Lessons from research on successful children. *American Psychologist*, *53*, 205–220.

Maulana, R., Opdenakker, M. C., & Bosker, R. (2014). Teacher–student interpersonal relationships do change and affect academic motivation: A multilevel growth curve modelling. *British Journal of Educational Psychology*, *84*(3), 459–482.

15348687, 2021, 177, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/cad.20415, Wiley Online Library on [17/12/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Moffitt, U., Juang, L. P., & Syed, M. (2019). "We don't do that in Germany!" A critical race theory examination of Turkish heritage young adults' school experiences. *Ethnicities*, *19*(5), 830–857.

Moos, R. H., & Trickett, E. J. (1987). *Classroom environment scale manual*. Palo Alto, CA: Consulting Psychologists Press.

Motti-Stefanidi, F., & Asendorpf, J. B. (2012). Perceived discrimination of immigrant adolescents in Greece. How does group discrimination translate into personal discrimination? *European Psychologist*, *17*, 93–104.

Motti-Stefanidi, F., & Masten, A. S. (2013). School success and school engagement of immigrant children and adolescents. *European Psychologist*, *18*, 126–135.

Muthén, L. K., & Muthén, B. O. (1998–2011). *Mplus user's guide*. Muthén & Muthén.

OECD (2019). *International migration outlook 2019*. OECD Publishing.

Oppedal, B., Røysamb, E., & Heyerdahl, S. (2005). Ethnic group, acculturation, and psychiatric problems in young immigrants. *Journal of Child Psychology and Psychiatry*, *46*(6), 646–660.

Pianta, R. C., Hamre, B. K., & Allen, J. P. (2012). Teacher-student relationships and engagement: Conceptualizing, measuring, and improving the capacity of classroom interactions. In S. L. Christenson (Ed.), *Handbook of research on student engagement* (pp. 365–386). Springer.

Price, M., Hill, N. E., Liang, B., & Perella, J. (2019). Teacher relationships and adolescents experiencing identity-based victimization: What matters for whom among stigmatized adolescents. *School Mental Health*, *11*(4), 790–806.

Quillian, L. (1995). Prejudice as a response to perceived group threat: Population composition and anti-immigrant and racial prejudice in Europe. *American Sociological Review*, *60*(4), 586–611.

Reyes, M. R., Brackett, M. A., Rivers, S. E., White, M., & Salovey, P. (2012). Classroom emotional climate, student engagement, and academic achievement. *Journal of Educational Psychology*, *104*, 700–712.

Roorda, D. L., Koomen, H. M., Spilt, J. L., & Oort, F. J. (2011). The influence of affective teacher–student relationships on students' school engagement and achievement: A meta-analytic approach. *Review of Educational Research*, *81*(4), 493–529.

Rosenberg, M. (1965). *Society and the adolescent self-image*. Princeton, NJ: Princeton University Press.

Rosseel, Y. (2012). lavaan: An R package for structural equation modeling. *Journal of Statistical Software*, *48*, 1–36.

Sabol, T. J., & Pianta, R. C. (2012). Recent trends in research on teacher–child relationships. *Attachment & Human Development*, *14*(3), 213–231.

Schachner, M. K., Juang, L., Moffitt, U., & Van de Vijver, F. J. R. (2018). Schools as acculturative and developmental context for youth of immigrant and refugee background. *European Psychologist*, *23*, 44–56.

Scherer, R., & Nilsen, T. (2016). The relations among school climate, instructional quality, and achievement motivation in Mathematics. In Nilsen, T. & Gustafsson, J.-E. (Eds.), *Teacher quality, instructional quality and student outcome. Relationships across countries, cohorts and time*. Springer International Publishing.

Schmitt, M. T., Branscombe, N. R., Postmes, T., & Garcia, A. (2014). The consequences of perceived discrimination for psychological well-being: A meta-analytic review. *Psychological Bulletin*, *140*, 921–948.

Schönbächler, M. T. (2008). *Klassenmanagement: Situative Gegebenheiten und personale Faktoren in Lehrpersonen-und Schülerperspektive*. Haupt Verlag AG.

Southwick, S. M., Bonanno, G. A., Mastern, A. S., Panter-Brick, C., & Yehuda, R. (2014). Resilience definitions, theory, and challenges: Interdisciplinary perspectives. *European Journal of Psychotraumatology*, *5*(1), 253–238.

Spilt, J. L., Koomen, H. M., & Jak, S. (2012). Are boys better off with male and girls with female teachers? A multilevel investigation of measurement invariance and gender match in teacher–student relationship quality. *Journal of School Psychology*, *50*(3), 363–378.

Statistiches Bundesamt (2016). *Hauptberufliches Bildungspersonal innerhalb und außerhalb früher Bildung, Schulen, und Hoschulen des Mikrozensus 2015*. Destatis.

Stevens, G. W., & Thijs, J. (2018). Perceived group discrimination and psychological well-being in ethnic minority adolescents. *Journal of Applied Social Psychology*, *48*, 559–570.

SVR-Forschungsbereich. (2018). *Wo kommen Sie eigentlich ursprünglich her?" Diskriminierungserfahrungen und phänotypische Differenz in Deutschland ["Where are you really from?" Discrimination experiences and phenotypical difference in Germany.]*. https://www.svr-migration.de/publikationen/

Taylor, D. M., Wright, S. C., Moghaddam, F. M., & Lalonde, R. N. (1990). The personal/group discrimination discrepancy: Perceiving my group, but not myself, to be a target for discrimination. *Personality and Social Psychology Bulletin*, *16*, 254–262.

Taylor, D. M., Wright, S. C., & Porter, L. E. (1994). Dimension of perceived discrimination: The personal/group discrimination discrepancy. In M. P. Zanna & M. J. Olson (Eds.), *The psychology of prejudice: The Ontario symposium* (pp. 233–256). Hillsdale.

Thapa, A., Cohen, J., Guffey, S., & Higgins-D'Alessandro, A. (2013). A review of school climate research. *Review of Educational Research*, *83*, 357–385.

Tucci, I., Eisnecker, P. P., & Brücker, H. (2014). *Diskriminierungserfahrungen und soziale Integration: Wie zufrieden sind Migranten mit ihrem Leben? [Experiences with discrimination and social integration: How satisfied are migrants with their life?.]* (No.21.4/2014). IAB-Kurbereicht.

15348687, 2021, 177, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/cad.20415, Wiley Online Library on [17/12/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Umaña-Taylor, A. J. (2016). A post-racial society in which ethnic-racial discrimination still exists and has significant consequences for youths' adjustment. *Current Directions in Psychological Science*, *25*(2), 111–118.

Umaña-Taylor, A. J., & Updegraff, K. A. (2007). Latino adolescents' mental health: Exploring the interrelations among discrimination, ethnic identity, cultural orientation, self-esteem, and depressive symptoms. *Journal of Adolescence*, *30*(4), 549–567.

Uslucan, H. H. (2017). Diskriminierungserfahrungen türkeistämmiger Zuwanderer_innen. In Fereidooni K., El, M. (Ed.) *Rassismuskritik und Widerstandsformen* (pp. 129–141). Springer.

Verkuyten, M. (1998). Perceived discrimination and self-esteem among ethnic minority adolescents. *The Journal of Social Psychology*, *138*(4), 479–493.

Verkuyten, M., Thijs, J., & Gharaei, N. (2019). Discrimination and academic (dis) engagement of ethnic-racial minority students: A social identity threat perspective. *Social Psychology of Education*, *22*(2), 267–290.

Vietze, J., Moffitt, U., Schwarzenthal, M., & Civitillo, S. (2021). *Beyond 'immigrant background': How to select relevant, social justice oriented, and feasible social categories in educational research*. Manuscript submitted for publication.

Wentzel, K. R. (2002). Are effective teachers like good parents? Teaching styles and student adjustment in early adolescence. *Child Development*, *73*, 287–301.

Wong, C. A., Eccles, J. S., & Sameroff, A. (2003). The influence of ethnic discrimination and ethnic identification on African American adolescents' school and socioemotional adjustment. *Journal of Personality*, *71*, 1197–1123.

Yip, T., Cheon, Y. M., & Erhardt, A. (2020). Applying experience-sampling methods to investigate the impact of school diversity on youth development in multicultural context. In P. F. Titzmann & Jugert, P. (Eds.), *Youth in Superdiverse Societies: Growing up with globalization, diversity, and acculturation* (pp. 111–128). Routledge.

Yip, T., Wang, Y., Mootoo, C., & Mirpuri, S. (2019). Moderating the association between discrimination and adjustment: A meta-analysis of ethnic/racial identity. *Developmental Psychology*, *55*(6), 1274–1298.

Zeiders, K. H., Umana-Taylor, A. J., & Derlan, C. L. (2013). Trajectories of depressive symptoms and self-esteem in Latino youths: Examining the role of gender and perceived discrimination. *Developmental Psychology*, *49*(5), 951.

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